Adapting Vision Foundation Models for Robust Cloud Segmentation in Remote Sensing Images

Xuechao Zou, Shun Zhang, Kai Li, *Student Member, IEEE*, Shiying Wang, Junliang Xing, *Senior Member, IEEE*, Lei Jin, Congyan Lang, *Member, IEEE*, and Pin Tao, *Member, IEEE*

***Abstract*—Cloud segmentation is a critical challenge in remote sensing image interpretation, as its accuracy directly impacts the effectiveness of subsequent data processing and analysis. Recently, vision foundation models (VFMs) have demonstrated powerful generalization capabilities across various visual tasks. In this article, we present a parameter-efficient adaptive approach, termed Cloud-Adapter, designed to enhance the accuracy and robustness of cloud segmentation. Our method leverages a VFM pretrained on general domain data, which remains frozen, eliminating the need for additional training. Cloud-Adapter incorporates a lightweight spatial perception module (SPM) that initially utilizes a convolutional neural network (ConvNet) to extract dense spatial representations. These multiscale features are then aggregated and serve as contextual inputs to an adapting module (AM), which modulates the frozen Transformer layers within the VFM. The experimental results demonstrate that the Cloud-Adapter approach, utilizing only 0.6% of the trainable parameters of the frozen backbone, achieves substantial performance gains. Cloud-Adapter consistently achieves state-of-the-art performance across various cloud segmentation datasets from multiple satellite sources, sensor series, data processing levels, land cover scenarios, and annotation granularities. Code and model checkpoints are available at https://xavierjiezou.github.io/Cloud-Adapter/**

***Index Terms*—Cloud segmentation, domain adaptation, fine-tuning, remote sensing image processing.**

Received 23 November 2024; revised 31 May 2025 and 18 July 2025; accepted 6 August 2025. Date of publication 11 August 2025; date of current version 20 August 2025. This work was supported in part by Beijing Natural Science-Xiaomi Innovation Joint Foundation under Grant L253007, in part by the National Natural Science Foundation of China under Grant 62402031 and Grant 62222606, and in part by Beijing Nova Program under Grant 20240484620. *(Xuechao Zou, Shun Zhang, and Kai Li contributed equally to this work.) (Corresponding authors: Congyan Lang; Pin Tao.)*

Xuechao Zou and Congyan Lang are with the School of Computer Science and Technology, Beijing Jiaotong University, Beijing 100044, China, and also with the Key Laboratory of Big Data and Artificial Intelligence in Transportation, Ministry of Education, Beijing 100044, China (e-mail: xuechaozou@foxmail.com; cylang@bjtu.edu.cn).

Shun Zhang and Shiying Wang are with the School of Computer Technology and Application, Qinghai University, Xining 810016, China (e-mail: xiaoshun3238@gmail.com; wangshiying.qhu@foxmail.com).

Kai Li is with the Department of Computer Science and Technology, Tsinghua University, Beijing 100084, China (e-mail: tsinghua.kaili@gmail.com).

Junliang Xing and Pin Tao are with the Department of Computer Science and Technology, Tsinghua University, Beijing 100084, China, and also with the Key Laboratory of Pervasive Computing, Ministry of Education, Beijing 100084, China (e-mail: jlxing@tsinghua.edu.cn; taopin@tsinghua.edu.cn).

Lei Jin is with the School of Electronic Engineering, Beijing University of Posts and Telecommunications, Beijing 100084, China (e-mail: jinlei@bupt.edu.cn).

Digital Object Identifier 10.1109/TGRS.2025.3597410

I. Introduction

CLOUD segmentation, also known as cloud detection, plays a critical role in the interpretation of remote sensing images, with wide applications in climate monitoring [1], [2], environmental protection, disaster warning, and land use [3]. The variations and characteristics of clouds have a significant influence on the quality of remote sensing images and the subsequent analysis results. Therefore, accurate cloud segmentation is essential for extracting meaningful geospatial information and ensuring reliable data processing. However, due to the diversity of cloud shapes, textures, and sizes, coupled with the complexity of imaging conditions, traditional cloud segmentation methods [4], [5], [6], [7] often fail to achieve stable, high-accuracy results in dynamic remote sensing images. Consequently, developing a more robust and adaptable cloud segmentation method has become a key research focus in remote sensing.

With the rapid development of deep learning, convolutional neural network (ConvNet)-based cloud segmentation methods [8], [9], [10], [11], [12], [13], [14] have made remarkable progress, particularly due to their ability to automatically extract high-level features that enhance the accuracy of cloud shape, edge detection, and texture analysis. For instance, MFCNN [10] combines multiscale convolutional features to enable fine-grained cloud detection. At the same time, DABNet [14] introduces the deformable context feature pyramid module to improve the adaptive modeling capability of multiscale features for high-accuracy cloud detection. Despite these advancements, many methods still face challenges, including strong dependence on training data, low model generalization, and limited adaptability to diverse remote sensing scenes.

Meanwhile, vision foundation models (VFMs) [15], [16], [17], [18], [19], which learn task-agnostic representations from large-scale datasets [20], have demonstrated impressive performance across various applications, such as image classification [21], object detection [22], and segmentation [23], [24], [25]. Transformer architectures, such as ViT [15] and Swin Transformer [16], have become dominant in the VFM landscape, demonstrating superior performance. However, the potential of VFMs for cloud segmentation remains largely unexplored.

Existing cloud segmentation methods [26], [27] primarily rely on complex neural network architectures, most of which require extensive retraining or fine-tuning on large labeled datasets. Although these methods achieve good performance,

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:19:28 UTC from IEEE Xplore. Restrictions apply.

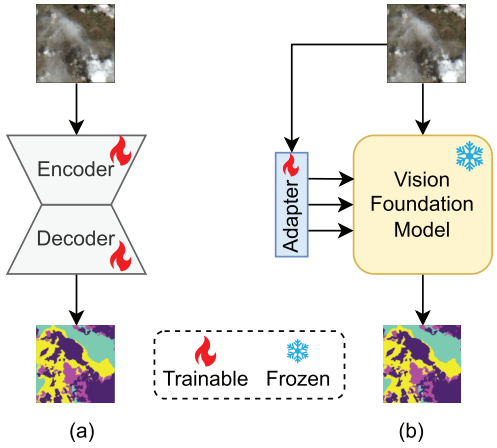


Fig. 1. Comparison between (a) previous method and (b) our Cloud-Adapter approach. Unlike (a), where the entire network (e.g., a typical encoder–decoder architecture) is fully trainable, resulting in a large number of parameters and an increased risk of overfitting, (b) our Cloud-Adapter approach leverages a frozen VFM combined with a lightweight, trainable adapter. This design preserves generalization and adaptability, enabling efficient learning for cloud segmentation tasks. The frozen VFM extracts generic visual features, while the parameter-efficient adapter facilitates effective transfer learning on diverse remote sensing scenes.

they suffer from several limitations. First, they are computationally expensive and require a large amount of labeled data, which is often impractical for real-world applications. Second, traditional models are prone to overfitting, particularly when dealing with small datasets or high-resolution remote sensing images. Third, most existing methods [12], [13], [26], [27], [28], [29], [30] have barely leveraged the power of VFM for cloud segmentation, missing the opportunity to exploit their robust transfer learning capabilities with minimal additional parameters.

To address these challenges, we propose a novel cloud segmentation method based on VFMs named Cloud-Adapter (see Fig. 1). This approach freezes the pretrained VFM backbone and incorporates a lightweight adaptive module, requiring minimal additional parameters to enhance cloud segmentation accuracy and robustness. Specifically, we design a spatial perception module (SPM) that employs convolutional neural networks (ConvNets) to extract dense spatial features and aggregate them into multiscale contextual information. This information is then passed into the adaptive module, which modulates the frozen Transformer layers of the VFM to enable efficient cloud segmentation. The experimental results demonstrate that Cloud-Adapter achieves significant performance improvements with only 0.6% of the trainable parameters of the backbone across various popular datasets.

The contributions of this work are summarized as follows.

1) We propose Cloud-Adapter, a simple but powerful parameter-efficient fine-tuning method for cloud segmentation that freezes the backbone of VFMs and fine-tunes only a small fraction of the frozen parameters.
2) We introduce two lightweight components: the SPM and the adapting module (AM), which modulate the features of the VFM, enabling adaptation from a general domain to a remote sensing domain.
3) We demonstrate that our method consistently achieves state-of-the-art performance on four datasets (six subsets), encompassing multiple satellite sources, sensor types, and land cover scenarios, thereby highlighting its robustness in remote sensing cloud segmentation tasks.

II. Related Work

A. Cloud Segmentation

Early cloud segmentation methods primarily relied on the spectral and statistical features of images, using techniques like thresholding, multispectral fusion, and edge detection to distinguish clouds from other surface elements. For instance, CFMask [6], [7] employs decision trees to label pixels and projects estimated cloud heights onto the ground to generate a cloud shadow mask. HOT [4], [5] introduces a transformation optimized for detecting cloud and haze distributions in Landsat images, amplifying haze variations via spectral sensitivity.

However, these traditional methods are highly sensitive to cloud shapes, lighting conditions, and atmospheric environments, resulting in suboptimal performance in complex remote sensing scenarios. With advancements in deep learning, ConvNet-based cloud segmentation methods have gained significant attention due to their ability to automatically extract high-level features, enhancing the accuracy of cloud shape, texture, and edge detection. Examples of innovative model architectures include MSCFF [8], which combines multiscale convolutional features to facilitate cloud detection. DBNet [27] employs a hybrid model that integrates Transformers and ConvNets to capture both semantic and spatial details, thereby reducing false detections and omissions in cloud segmentation. HRCloudNet [30] adopts a hierarchical integration approach to preserve cloud texture in high-resolution images, while MCD-Net [26] achieves effective thin cloud segmentation through multiscale feature fusion and auxiliary cloud removal.

For models designed for specific scenarios and training needs, CDNetv1 [12] leverages a feature pyramid and a boundary refinement block to extract cloud masks from low-resolution thumbnails. CDNetv2 [13] improves performance in mixed cloud-snow scenes using adaptive feature fusion and high-level semantic guidance. SCNN [28] applies a shallow ConvNet with filters of varying sizes to minimize model training costs. Additionally, TransMCD [31] uses pseudo-labels derived from spectral features, enabling weakly supervised cloud detection and enhancing model generalization. Despite these advancements [32], [33], [34], [35], achieving robust segmentation performance across diverse remote sensing images remains a significant challenge. Furthermore, most methods continue to rely on traditional network architectures, such as UNet [36], ResNet [37], FPN [38], and DeepLab [39], with limited exploration of VFMs for cloud segmentation.

B. Vision Foundation Models

Learning task-agnostic pretrained representations [40], [41] has become a standard practice in natural language processing (NLP). Inspired by this paradigm, substantial progress has been achieved in exploring VFMs. Typically, these

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:19:28 UTC from IEEE Xplore. Restrictions apply.

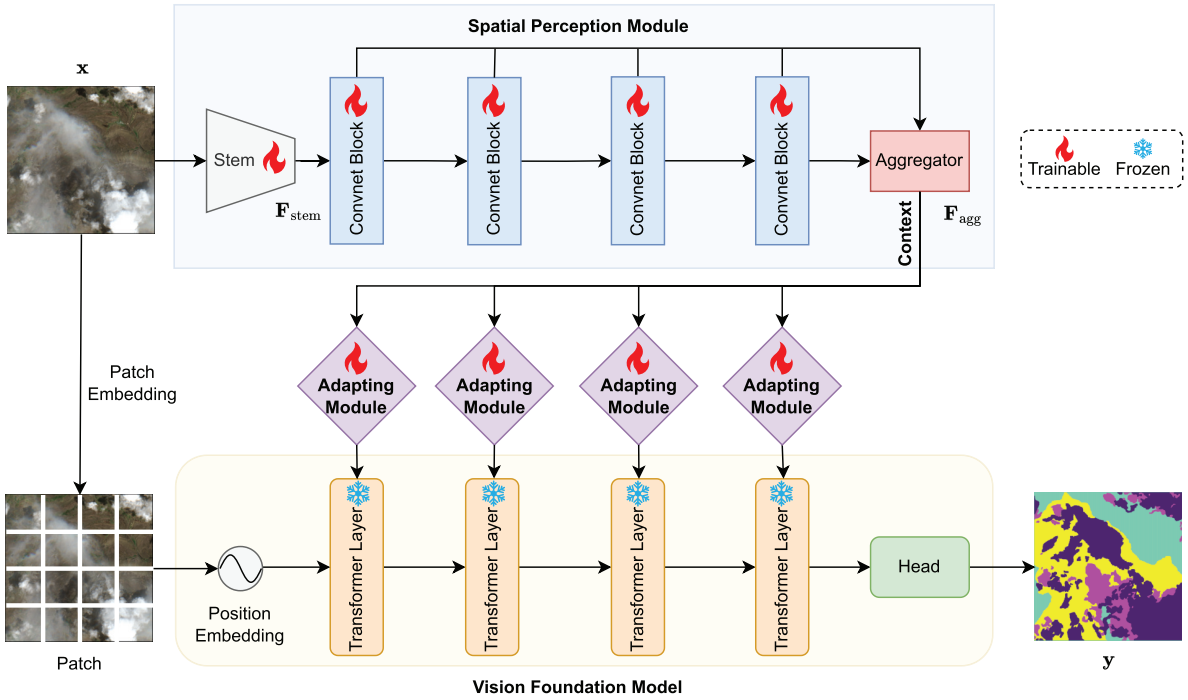


Fig. 2. Detailed network architecture of the proposed Cloud-Adapter method, consisting of the spatial perception and adapting modules. The SPM uses ConvNet blocks to extract dense spatial features, which are aggregated into a multiscale context and fed to the AM. The AM modulates the frozen Transformer layers in the VFM.

models are trained on large-scale datasets to learn general-purpose representations, achieving remarkable performance across diverse applications, including image classification, object detection, and segmentation. Transformer architectures [15], [16] dominate the landscape of VFMs, demonstrating superior performance. CLIP [42] aligns text and image modalities in latent space through contrastive learning. MAE [17] learns robust representations via self-supervised masked modeling. SAM [18] incorporates interactive prompts to capture general concepts of objects, while DINOv2 [19] employs self-distillation to learn generic semantic features. These models effectively capture complex visual patterns and exhibit strong capabilities for few-shot and even zero-shot transfer, allowing them to adapt well to new tasks. Given the diversity and complexity of remote sensing images, exploring VFMs with strong generalization is essential for achieving robust cloud segmentation.

C. Transfer Learning and Domain Adaptation

In resource-constrained environments or when dealing with large-scale data, training models from scratch is often impractical and costly. Transfer learning [43] enables the adaptation of models trained on large datasets to specific tasks. This approach typically involves fine-tuning pretrained models on smaller, domain-specific datasets, allowing the model to retain general knowledge while learning domain-specific characteristics. Domain adaptation [44] further enhances this process by aligning the feature distributions between the source domain (where the model was initially trained) and the target domain (the new application environment), thus maximizing task performance.

Adapters exemplify the integration of transfer learning and domain adaptation, offering a generally plug-and-play approach across various tasks. Initially introduced in natural language processing (NLP) [45], chapters have undergone rapid evolution. In NLP tasks, LoRA [46] injects trainable low-rank decomposition matrices into each layer of the Transformer architecture, enabling large language models to adapt to specific domains. In multimodal generation tasks, IP-Adapter [47] is an effective adapter for implementing image prompting functions in pretrained text-to-image diffusion models. T2I-Adapter [48] introduces a lightweight adapter to modulate the intensity of internal knowledge and external control signals in text-to-image models. For visual segmentation tasks, ViT-Adapter [49] integrates image-related inductive biases into pretrained ViT models, improving dense prediction performance. SAM-Adapter [50] introduces a simple adapter that incorporates domain-specific information or visual prompts into SAM [18], enhancing performance in specific scenarios. Therefore, developing an efficient adapter shows excellent potential for improving the robustness of cloud segmentation by leveraging the strengths of VFMs.

III. Method

In this section, we introduce a simple but powerful adaptive method called Cloud-Adapter (as shown in Fig. 2) to achieve fine-tuning of cloud segmentation by freezing the weights of the VFM. The method consists of two key components: the SPM and the adapting modules. The spatial perception module extracts multiscale spatial features using multiple ConvNet blocks and then aggregates them into a dense context. The adapting modules modulate the frozen VFM layers with the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:19:28 UTC from IEEE Xplore. Restrictions apply.

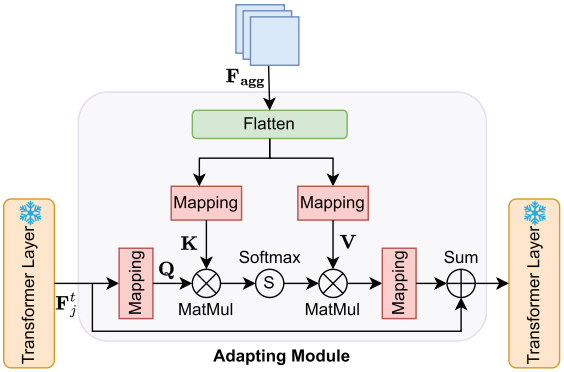


Fig. 3. Schematic of the proposed AM.

context, enabling effective transfer learning for cloud segmentation.

A. Spatial Perception Module

The SPM extracts multiscale spatial features from the input remote sensing images. First, the input image goes through a stem module for preliminary feature extraction. The stem consists of two depthwise separable convolutions [51]. The first convolution uses a 1×1 kernel to map the input image's channels to a larger feature space, primarily for channel-wise information fusion. The second convolution uses a 3×3 kernel to extract spatial features and capture local spatial information.

Let the input image be $\mathbf{x} \in \mathbb{R}^{C \times H \times W}$, where C is the number of channels, and H and W are the height and width of the image. The output feature map after stem is $\mathbf{F}_{\text{stem}} \in \mathbb{R}^{C' \times H \times W}$, where C' is the mapped number of channels, and H and W are the feature map dimensions, which remain the same as the input image.

After the stem, the input features are processed by multiple ConvNet blocks, each of which extracts spatial features at different scales, resulting in a series of feature maps $\mathbf{F}_i^c \in \mathbb{R}^{C_i \times H_i \times W_i}$, where i denotes different scales. To aggregate these multiscale feature maps into a unified scale, we use an aggregator, which applies adaptive average pooling to normalize the feature maps to the same size. The aggregator is a parameter-free operation, meaning that it does not contain any trainable parameters, making it a zero-parameter operation.

The aggregation of multiscale features is expressed as

$$\mathbf{F}_{\text{agg}} = \sum_{i=1}^{k} \mathcal{P}_{\text{avg}}\left(\mathbf{F}_i^c\right) \tag{1}$$

where $\mathbf{F}_i^c$ are the feature maps at different scales and $\mathcal{P}_{\text{avg}}$ is the adaptive average pooling operation that pools the feature maps to the smallest size of the final feature map $\mathbf{F}_k^c$. Additionally, $k \in \{1, 2, \ldots, \lfloor \log_2(\min(H_0, W_0)) \rfloor\}$ refers to the number of ConvNet blocks, where each block has a convolution with the stride of 2, achieving a downsampling effect by squeezing the spatial dimensions at each step. Using the aggregator, we integrate multiscale hierarchical representations into a single dense prior $\mathbf{F}_{\text{agg}} \in \mathbb{R}^{C_k \times H_k \times W_k}$.

B. Adapting Module

The AM, as referred to in Fig. 3, utilizes the cross-attention mechanisms [52], [53], [54], [55] to adjust the frozen VFM's Transformer layers. Specifically, it takes the context $\mathbf{F}_{\text{agg}}$ from the SPM and interacts with the frozen VFM features, thereby modulating the output of the VFM. The cross-attention mechanism calculates the relationships between aggregated dense features and the features extracted by the VFM, producing the adapted features. The queries, keys, and values are computed as follows:

$$\mathbf{Q} = \mathcal{M}_q\left(\mathbf{F}_j^t\right), \quad \mathbf{K} = \mathcal{M}_k\left(\mathbf{F}_{\text{agg}}\right), \quad \mathbf{V} = \mathcal{M}_v\left(\mathbf{F}_{\text{agg}}\right) \tag{2}$$

where $\mathbf{Q} \in \mathbb{R}^d$, $\mathbf{K} \in \mathbb{R}^d$, and $\mathbf{V} \in \mathbb{R}^d$ are the query, key, and value matrices, respectively. $\mathcal{M}_q$, $\mathcal{M}_k$, and $\mathcal{M}_v$ denote learnable linear projection functions for generating the respective matrices. $\mathbf{F}_j^t$ represents the robust feature extracted from the frozen VFM Transformer layers, where j indicates the index. Then, the cross-attention computation is expressed by

$$\mathbf{S} = \text{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d}}\right) \tag{3}$$

$$\mathbf{O} = \mathbf{S} \cdot \mathbf{V} \tag{4}$$

$$\mathbf{F}_j^{\text{att}} = \mathcal{M}\left(\mathbf{O}\right) + \mathbf{F}_j^t \tag{5}$$

where the Softmax operation computes the attention weights $\mathbf{S}$, which are then multiplied with the value matrix $\mathbf{V}$ to obtain the output $\mathbf{O}$. Finally, after the mapping operation (denoted as $\mathcal{M}$), the adapted features $\mathbf{F}_j^{\text{att}}$ are obtained with a residual.

The mapping operation is performed via a multilayer perceptron (MLP) with the low-rank [46] design, without any activation, aimed at achieving feature projection with minimal parameter cost. The mapping process can be expressed as

$$\mathcal{M}\left(\mathbf{z}\right) = \mathbf{W}_2 \cdot \left(\mathbf{W}_1 \cdot \mathbf{z}\right) \tag{6}$$

where $\mathbf{W}_1 \in \mathbb{R}^{d \times r}$ and $\mathbf{W}_2 \in \mathbb{R}^{r \times d}$ are the weight matrices of the fully connected layers, $\mathbf{z}$ is the input feature, and r is the rank for dimensionality reduction. The smaller the value of r, the fewer trainable parameters the AM contains.

IV. Experiments

A. Datasets

We conducted experiments on four popular cloud segmentation datasets, comprising six subsets, from different satellite sources: CloudSEN12_High [56], L8_Biome [7], GF12MS_WHU [31], and HRC_WHU [8], as summarized in Table I. Specifically, CloudSEN12_High contains two data processing levels, L1C and L2A, whereas GF12MS_WHU consists of data captured by two satellite sensors, GF1 and GF2. Notably, in typical binary cloud segmentation datasets, thin and thick clouds are grouped into a single category, "cloud," while cloud shadows are excluded from the annotations.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:19:28 UTC from IEEE Xplore. Restrictions apply.

TABLE I
SUMMARY OF INFORMATION FOR FOUR POPULAR CLOUD SEGMENTATION DATASETS. "-" INDICATES NONEXISTENCE OR NOT MENTIONED

Dataset Name	Satellite Source	Ground Resolution	Training Set	Validation Set	Testing Set	Patch Size	Classes
CloudSEN12_High [56]	Sentinel 2	10m	8,490	535	975	512×512	4
L8_Biome [7]	Landsat 8	30m	7,931	2,643	2,643	512×512	4
GF12MS_WHU [31]	Gaofen 1, 2	8m & 4m	20,697	11,645	-	256×256	2
HRC_WHU [8]	Google Earth	0.5m-15m	120	-	30	256×256	2

1) CloudSEN12_High [56]: CloudSEN12 is a global dataset designed for the semantic understanding of clouds and their shadows in Sentinel-2 imagery. It addresses the long-standing challenge of accurately characterizing clouds and cloud shadows in the Earth observation community. This dataset comprises 49400 image patches, including Sentinel-2 Level-1C and Level-2A multispectral data, Sentinel-1 synthetic aperture radar data, auxiliary remote sensing products, and meticulously crafted annotations identifying thick clouds, thin clouds, and cloud shadows. CloudSEN12 stands out due to its diverse annotations, scene variability, broad geographic coverage, and metadata complexity, making it an essential resource for benchmarking and improving cloud detection methodologies. Here, we select the high-quality version with four classes of fine-grained annotations for our experiments.

2) L8_Biome [7]: The L8_Biome dataset is designed to facilitate the comparison and validation of cloud detection algorithms for operational Landsat data products. This dataset encompasses various scenario types: Urban, Barren, Forest, Shrubland, Grass/Cropland, Snow/Ice, Wetlands, and Water. Such a variety of scene types enables models to learn more diverse features, handle complex surface backgrounds, and enhance the accuracy of cloud removal tasks across different terrains. The dataset includes manually generated cloud masks to support the validation of cloud cover assessment algorithms, enabling precise computation of cloud cover percentages for each scene. These characteristics make the dataset an essential resource for cloud segmentation. Here, we cropped each tile into 512 × 512 patches and removed images that did not contain any targets of interest. The remaining data were then split into training, validation, and test sets in a ratio of 6:2:2.

3) GF12MS_WHU [31]: The GFMS_WHU dataset consists of 8-m Gaofen1-MS and 4-m Gaofen2-MS images. Gaofen1 and Gaofen2 belong to the same satellite series but are equipped with different sensors, offering varying spatial resolutions to enhance the model's generalization capability. These images were collected from diverse regions of China between June 2014 and December 2020, improving the dataset's ability to support robust model training. To standardize input dimensions, all images were uniformly cropped and padded to a size of 256 × 256 patches. The dataset is split into a training set and a test set with a ratio of 7:3, providing a balanced framework for model development and evaluation.

4) HRC_WHU [8]: HRC_WHU is a high-resolution cloud cover validation dataset. It comprises 150 high-resolution images acquired from Google Earth with RGB channels and resolutions varying from 0.5 to 15 m. These images have been expertly digitized with two mask annotations: clear sky and clouds. Here, we uniformly cut to a 256 × 256 patch size, 120 for training, and 30 for testing.

B. Implementation Details

1) Training Settings: All experiments were conducted on a workstation with NVIDIA GeForce RTX 3090 GPUs. The number of ConvNet blocks (k) was set to 4. We employed a custom learning rate schedule to optimize model performance. The schedule began with a warm-up phase, utilizing a linear scheduler that increased the learning rate from 1×10^{-6} over the first 1000 iterations. This was followed by a PolyLR scheduler with an eta minimum of 0.0 and a power of 0.9, applied from iteration 1000 to 40000. The training loop was configured to run for a maximum of 40000 iterations, with validation conducted every 4000 steps. To ensure reproducibility, we set the random seed to 42 and used the AdamW optimizer [57] with an initial learning rate of 1×10^{-4} and a weight decay of 0.05. The batch size during training was set to 4, and parameters were initialized using the Kaiming normal [58]. The segmentation head used in this study was the widely used Mask2Former [59] to generate masks.

2) Evaluation Metrics: To comprehensively evaluate cloud segmentation performance, we adapt several widely recognized metrics: intersection over union (IoU), accuracy (Acc), and Dice coefficient (Dice), all reported as percentages (%). Furthermore, we provided the proposed method's trainable parameters and their proportion relative to the frozen VFM backbone, denoted as "*Params." The IoU quantifies the overlap between the predicted and the ground truth masks

$$\mathrm{IoU}_i = \frac{|\mathrm{Pred}_i \cap \mathrm{GT}_i|}{|\mathrm{Pred}_i \cup \mathrm{GT}_i|} = \frac{\mathrm{TP}_i}{\mathrm{TP}_i + \mathrm{FP}_i + \mathrm{FN}_i} \tag{7}$$

where $\mathrm{TP_i}$, $\mathrm{FP_i}$, and $\mathrm{FN_i}$ represent the true positives, false positives, and false negatives for class i, respectively. The Acc represents the pixel-wise classification accuracy

$$\mathrm{Acc}_i = \frac{\mathrm{TP}_i + \mathrm{TN}_i}{\mathrm{TP}_i + \mathrm{TN}_i + \mathrm{FP}_i + \mathrm{FN}_i} \tag{8}$$

where $\mathrm{TN_i}$ represents the true negatives for class i. The Dice measures the similarity between the predicted and ground truth masks, and it is equivalent to the $F1$-score metric

$$\mathrm{Dice}_i = \frac{2\,|\mathrm{Pred}_i \cap \mathrm{GT}_i|}{|\mathrm{Pred}_i| + |\mathrm{GT}_i|} = \frac{2\mathrm{TP}_i}{2\mathrm{TP}_i + \mathrm{FP}_i + \mathrm{FN}_i}. \tag{9}$$

The metrics mAcc, mIoU, and mDice represent the mean values computed for each class, offering insight into the model's performance at the class level. In contrast, the overall accuracy (aAcc) evaluates the model's classification performance across the entire dataset. These metrics are all positive indicators, with higher values signifying better performance, simplifying interpretation, and facilitating model comparison.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:19:28 UTC from IEEE Xplore. Restrictions apply.

TABLE II

CLOUD SEGMENTATION PERFORMANCE COMPARISON OF DIFFERENT VFM'S BACKBONE WITH THEIR MULTIPLE VARIANTS

Backbone	Variant	mIoU	mAcc	aAcc	mDice	*Params (M)
SAM [18]	Base	72.53	83.29	89.46	83.22	0.92 (1.1%)
	Large	73.36	83.90	89.84	83.85	1.83 (0.6%)
	Huge	73.70	84.37	89.95	84.07	2.63 (**0.4%**)
DINOv2 [19]	Small	68.12	79.37	87.50	79.84	**0.77** (3.4%)
	Base	71.95	83.02	89.22	82.82	0.92 (1.1%)
	Large	**74.18**	**84.79**	**90.19**	**84.46**	1.82 (0.6%)

TABLE III

COMPARISON OF CLOUD SEGMENTATION PERFORMANCE AND MODEL PARAMETERS FOR DIFFERENT CONVNET ARCHITECTURES

ConvNet Block	mIoU	mAcc	aAcc	mDice	*Params (M)
Transformer-Like [60]	74.08	**84.95**	90.09	84.39	**1.81** (**0.6%**)
Pure-ConvNet [55]	**74.18**	84.79	**90.19**	**84.46**	1.82 (0.6%)

C. Ablation Studies and Analysis

This section presents a series of comprehensive ablation studies to evaluate the effectiveness of key components in our proposed model using the CloudSEN12_High_L1C dataset.

1) Variants of VFM's Backbone: To evaluate the impact of different VFMs and their variants on our method's performance, we conducted comprehensive experiments using SAM [18] and DINOv2 [19] as backbones. Table II compares three available model variants. Specifically, SAM offers [Base, Large, Huge], while DINOv2 provides [Small, Base, Large, Giant]. Thus, we chose the most comparable variants available for a fair evaluation.

We observed that SAM and DINOv2 exhibited consistent performance improvements as the model scale increased. Among the SAM variants, the Huge version performed best, achieving 73.70% mIoU, 84.37% mAcc, 89.95% aAcc, and 84.07% mDice. In contrast, DINOv2 (Large) demonstrated superior performance across all metrics, reaching 74.18% mIoU, 84.79% mAcc, 90.19% aAcc, and 84.46% mDice.

More importantly, DINOv2 (Large) was significantly more computationally efficient, requiring only 11 h and 8 min of training time and 14558M of GPU memory. In comparison, the SAM (Huge) variant took 19 h and 9 min to train, with GPU memory usage of 24892 MB, which was substantially higher than DINOv2 (Large).

These results indicate that DINOv2 (Large) outperforms SAM (Huge) across all metrics while being significantly more efficient in terms of training time and memory usage. Therefore, DINOv2 (Large) emerges as the optimal backbone for our method, striking an excellent balance between performance and computational efficiency for cloud segmentation.

2) ConvNet Architectures of SPM: We investigate the impact of different ConvNet architectures on the performance of our method, as shown in Table III. Specifically, we compare two types of ConvNet designs: a Transformer-like architecture inspired by [60] and a pure ConvNet structure from [55].

The Pure-ConvNet [55] architecture achieves slightly better overall performance with 74.18% mIoU, 90.19% aAcc, and 84.46% mDice, while the Transformer-Like [60] design shows marginally higher mAcc at 84.95%. Both architectures maintain similar parameter efficiency, with approximately 1.8 million parameters.

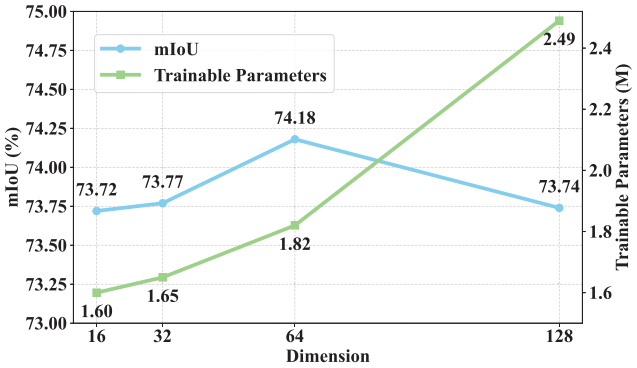


Fig. 4. Ablation study of different dimension settings on the interactive context extracted by the SPM.

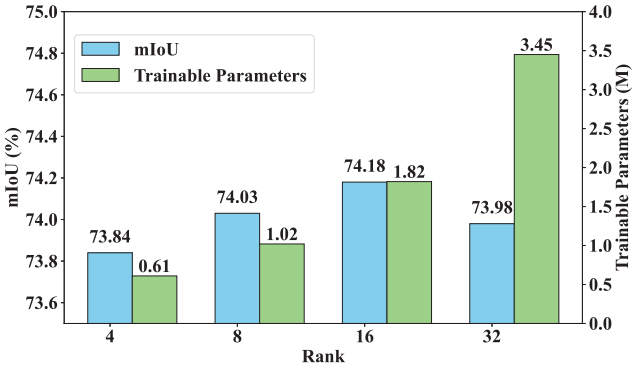


Fig. 5. Study of the low-rank MLP in the AM.

The comparable performance between these two architectures suggests that both designs are viable options for our method. However, considering the slightly higher performance in most metrics and the simpler structure, we adopt the Pure-ConvNet architecture as our default choice.

3) Dimension of Interactive Context: We conduct experiments to explore the impact of different dimension settings on the interactive context extracted by the SPM. As illustrated in Fig. 4, we evaluate four configurations: 16, 32, 64, and 128.

The results show that the model's performance does not necessarily improve with larger dimensions. Starting from dimension 16 with an mIoU of 73.72%, we observe a slight improvement to 73.77% when increasing to dimension 32. The peak performance is achieved at dimension 64 with an mIoU of 74.18%. However, further increasing the dimension to 128 leads to slight degradation (a reduction of 0.44%).

Meanwhile, the number of trainable parameters grows steadily with the dimension size. The model requires 1.60 M parameters at dimension 16, gradually increasing to 1.65 M at dimension 32, 1.82 M at dimension 64, and significantly jumping to 2.49 M at dimension 128. Based on these observations, we identify dimension 64 as the optimal choice, offering the best tradeoff between performance and efficiency.

4) Low-Rank MLP of AM: We investigate the influence of different rank settings on the performance and parameter efficiency of our model. The rank dimension controls the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:19:28 UTC from IEEE Xplore. Restrictions apply.

TABLE IV

CLOUD SEGMENTATION PERFORMANCE COMPARISON OF CONTEXT OUTPUT STRATEGIES FOR THE SPM

Feature Type	mIoU	mAcc	aAcc	mDice	*Params (M)
Single-Scale	74.05	84.45	90.19	84.34	1.82 (0.6%)
Multi-Scale	74.10	84.70	90.15	84.40	1.82 (0.6%)
Aggregated	**74.18**	**84.79**	**90.19**	**84.46**	1.82 (0.6%)

TABLE V

COMPARISON OF PERFORMANCE AT DIFFERENT INTERACTION FREQUENCIES (N) BETWEEN THE ADAPTER AND TRANSFORMER LAYERS

N	mIoU	mAcc	aAcc	mDice	*Params (M)
4	73.79	84.22	90.10	84.17	**0.47** (**0.2**%)
8	73.76	84.46	89.99	84.15	0.74 (0.2%)
12	73.75	84.46	89.99	84.13	1.02 (0.3%)
24	**74.18**	**84.79**	**90.19**	**84.46**	1.82 (0.6%)

size of the low-rank matrices used to approximate the weight updates in the MLP of the AM, thereby reducing the number of trainable parameters. As shown in Fig. 5, we examine four rank settings: 4, 8, 16, and 32.

The experimental results reveal an interesting relationship between rank size and model performance. With a rank of 4, the model achieves an mIoU of 73.84% while requiring only 0.61 M trainable parameters. Increasing the rank to eight improves the performance to 74.03% with 1.02 M parameters. The peak performance is reached at rank 16 with an mIoU of 74.18%, which requires 1.82 M parameters—further increasing the rank to 32 leads to a slight performance degradation. Based on these observations, we choose rank 16 as our optimal setting. This finding aligns with LoRA's design principle [46] of achieving efficient adaptation through low-rank approximation.

5) Importance of the Aggregator: We explore three context output strategies for our SPM: single-scale, multiscale, and aggregated approaches. Table IV presents the comparative results of these strategies while maintaining the exact parameter count of 1.82M (0.6% of total parameters).

The "Single-Scale" strategy, which directly utilizes the final feature output from the last ConvNet block, achieves baseline performance with 74.05% mIoU, 84.45% mAcc, and 84.34% mDice. The "Multiscale" approach, leveraging multiple scale features without fusion, shows slight improvements with 74.10% mIoU, 84.70% mAcc, and 84.40% mDice.

The "Aggregated" strategy, which fuses multiple scale features into a single output with parameter-free operation, demonstrates the best overall performance. It achieves 74.18% mIoU, 84.79% mAcc, and 84.46% mDice while maintaining the same aAcc (90.19%) as the "Single-Scale" approach. This superior performance can be attributed to the strategy's ability to effectively combine and utilize information from different feature scales. Based on these results, we adopt the "Aggregated" feature output strategy in our final model design, as it provides the most effective use of dense multiscale feature information without introducing additional parameters.

6) Interaction Frequency of AM: In this ablation study, we explore the effect of varying the interaction frequency (N) between the adapter and Transformer layers on the model's performance. The parameter N indicates how frequently the adapter interacts with the Transformer layers, with larger values corresponding to more frequent interactions. Table V presents the results of this experiment, evaluating different values of N: 4, 8, 12, and 24. Specifically, when N is set to 4, the adapter interacts with the Transformer layers at indices [0, 6, 12, 18]; "8" and "12" represent interactions at every third and second layer starting from index 0, respectively.

TABLE VI

STUDY OF EACH COMPONENT IN THE CLOUD-ADAPTER

Spatial Perception Module			Adapting Module	mIoU	*Params (M)
Stem	ConvNet Block	Aggregator			
×	×	×	×	73.10	**0.00**
✓	×	×	×	73.12	3e-4
✓	✓	×	×	73.20	0.20
✓	✓	✓	×	73.37	0.20
✓	✓	✓	✓	**74.18**	1.82

The results show that the best performance is achieved when $N = 24$, yielding 74.18% mIoU, 84.79% mAcc, 90.19% aAcc, and 84.46% mDice. This setting outperforms others, especially in mIoU and mAcc, indicating that increasing the interaction frequency enhances the model's ability to capture complex spatial dependencies. While lower N values (4, 8, and 12) maintain stable but suboptimal performance, the parameter increase with higher N remains minimal (only 0.2%–0.6%), due to the efficient LoRA-MLP design. These results suggest that allowing the adapter to interact with more Transformer layers improves performance, with $N = 24$ offering an effective balance between accuracy and efficiency.

7) Overall Components in the Cloud-Adapter: To evaluate the contribution of each component in our model, we conducted an ablation study by systematically removing or modifying specific modules. The results are presented in Table VI, showing the performance in terms of mIoU and parameter count for different configurations. Notably, in configurations without the AM, the features output by the SPM are concatenated with the features from the VFM backbone before being passed to the head for prediction.

In the baseline model, which utilizes only DINOv2 Large (without spatial perception and adaptation modules), features are directly passed to the head, achieving an mIoU of 73.10%. Adding the lightweight stem module slightly improved mIoU to 73.12% with a minimal parameter increase (3×10^{-4}M), by providing basic spatial awareness. Incorporating the stem and ConvNet Block further boosted performance to 73.20% (+0.20 M parameters), enabling better spatial feature extraction. Adding the aggregator module increased mIoU to 73.37% without adjusting parameters, demonstrating its effectiveness in multiscale feature fusion. Finally, the full model, including the AM, achieved the highest mIoU of 74.18% with 1.82 M parameters, highlighting the adapting module's key role in optimizing feature integration. The ablation study shows that each module contributes incrementally to improved segmentation performance.

8) Comprehensive Analysis of Efficiency–Performance Tradeoffs: Our parameter efficiency primarily stems from the carefully designed SPM, which leverages lightweight convnet

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:19:28 UTC from IEEE Xplore. Restrictions apply.

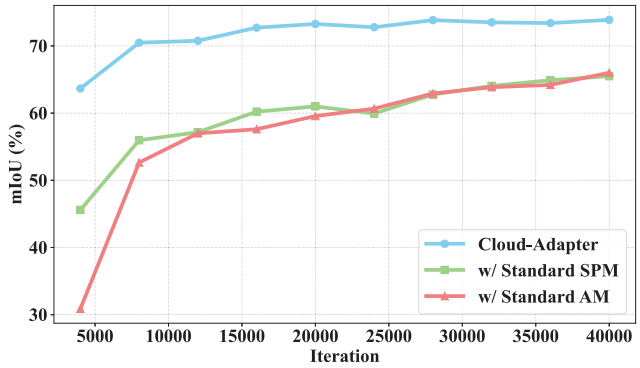


Fig. 6. Comparison of convergence speed between Cloud-Adapter and its standard-parameter variants.

TABLE VII

PERFORMANCE COMPARISON OF CLOUD-ADAPTER AND ITS STANDARD-PARAMETER VARIANTS UNDER CLEAN (LEFT) AND NOISY (RIGHT) CONDITIONS FOR ROBUSTNESS EVALUATION AND ANALYSIS

Method	mIoU	aAcc	*Params (M)
w/ Standard SPM	65.50 / 17.14	85.78 / 49.68	27.05 (8.2%)
w/ Standard AM	66.01 / 17.78	86.11 / 51.55	26.95 (8.1%)
Cloud-Adapter	**74.18 / 51.67**	**90.19 / 76.38**	**1.82 (0.6%)**

blocks to extract dense, multiscale contextual cues, as well as the use of a LoRA-MLP in the AM to enable efficient adaptation with minimal extra parameters. To comprehensively evaluate the tradeoffs between efficiency and performance, we conduct ablation studies by replacing the lightweight convnet blocks in the SPM with standard convolutional blocks [37], and replacing the LoRA-MLP in the AM with a standard MLP. As shown in Table VII, both standard-parameter variants suffer substantial drops in mIoU and aAcc under clean conditions, demonstrating the necessity of our lightweight modules for effective segmentation. Furthermore, since remote sensing imagery is often affected by various types of noise, we evaluate robustness by introducing Gaussian noise to the input images, following [61]. The results indicate that Cloud-Adapter maintains strong performance even in noisy scenarios, whereas the standard-parameter variants degrade sharply, approaching nearly random guessing. Additionally, convergence analysis (see Fig. 6) shows that Cloud-Adapter not only achieves higher final accuracy but also converges significantly faster than the standard-parameter variants. Overall, these results confirm that the parameter-efficient design of Cloud-Adapter, enabled by the SPM and AM, does not sacrifice segmentation accuracy, convergence speed, or robustness but instead achieves a favorable balance between lightweight design and strong performance.

D. Comparison With State-of-the-Art Methods

We compare the performance of our proposed Cloud-Adapter with several state-of-the-art cloud segmentation methods across different datasets: HRC_WHU, GF12MS_WHU_GF1, GF12MS_WHU_GF2, CloudSEN12_High_L1C, and CloudSEN12_High_L2A. These datasets include both binary (HRC_WHU, GF1, and GF2) and multiclass (L1C and L2A) cloud segmentation. The performance comparison, as shown in Tables VIII–XII, highlights Cloud-Adapter's robustness in both binary and multiclass cloud segmentation.

TABLE VIII

CLOUD SEGMENTATION PERFORMANCE COMPARISON OF DIFFERENT METHODS ON THE HRC_WHU DATASET

Method	aAcc	mIoU	mAcc	mDice
MCDNet [26]	75.14	53.50	68.91	67.96
SCNN [28]	74.51	57.22	81.27	72.31
KappaMask [29]	84.73	67.48	80.30	79.74
CDNetv2 [13]	89.71	76.75	87.46	86.46
DBNet [27]	90.11	77.78	88.80	87.17
CDNetv1 [12]	89.88	77.79	89.93	87.20
UNetMobv2 [56]	92.13	79.91	85.61	88.45
HRCloudNet [30]	92.93	83.44	92.39	90.79
Cloud-Adapter (Ours)	**94.50**	**89.05**	**93.74**	**94.19**

TABLE IX

CLOUD SEGMENTATION PERFORMANCE COMPARISON OF DIFFERENT METHODS ON THE GF12MS_WHU_GF1 DATASET

Method	aAcc	mIoU	mAcc	mDice
SCNN [28]	97.18	81.68	87.21	89.13
CDNetv1 [12]	96.81	81.82	92.75	89.27
CDNetv2 [13]	97.55	84.93	92.96	91.36
MCDNet [26]	97.55	85.16	93.97	91.51
UNetMobv2 [56]	98.82	91.71	93.99	95.53
DBNet [27]	98.73	91.36	95.19	95.33
HRCloudNet [30]	98.80	91.86	95.82	95.62
KappaMask [29]	98.91	92.42	95.05	95.94
Cloud-Adapter (Ours)	**98.92**	**92.55**	**95.99**	**96.02**

TABLE X

CLOUD SEGMENTATION PERFORMANCE COMPARISON OF DIFFERENT METHODS ON THE GF12MS_WHU_GF2 DATASET

Method	aAcc	mIoU	mAcc	mDice
KappaMask [29]	90.30	72.00	77.64	82.57
HRCloudNet [30]	91.46	75.57	80.95	85.29
SCNN [28]	91.99	76.99	82.06	86.32
CDNetv1 [12]	92.42	78.20	83.07	87.17
MCDNet [26]	92.30	78.36	83.95	87.31
DBNet [27]	92.61	78.68	83.39	87.50
CDNetv2 [13]	92.63	78.84	83.67	87.61
UNetMobv2 [56]	93.22	80.44	84.86	88.70
Cloud-Adapter (Ours)	**94.04**	**83.02**	**87.54**	**90.40**

1) Binary-Class Cloud Segmentation:

a) Quantitative comparison: Cloud-Adapter consistently performs well across all evaluation metrics (aAcc, mIoU, mAcc, and mDice) for binary-class cloud segmentation datasets. On the HRC_WHU dataset (Table VIII), Cloud-Adapter achieves the highest scores with an aAcc of 94.50%, mIoU of 89.05%, mAcc of 93.74%, and mDice of 94.19%, significantly outperforming the second-best method, HRCloudNet, which only achieves an mIoU of 83.44%. On the GF12MS_WHU_GF1 dataset (Table IX), Cloud-Adapter attains an aAcc of 98.92%, mIoU of 92.55%, mAcc of 95.99%, and mDice of 96.02%, surpassing KappaMask, which achieved a slightly lower mIoU of 92.42%. On the GF12MS_WHU_GF2 dataset (Table X), Cloud-Adapter maintains its leading performance with an aAcc of 94.04%, mIoU of 83.02%, mAcc of 87.54%, and mDice of 90.40%, showing its robustness in handling challenging satellite data.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:19:28 UTC from IEEE Xplore. Restrictions apply.

TABLE XI

CLOUD SEGMENTATION PERFORMANCE COMPARISON OF DIFFERENT METHODS ON THE CLOUDSEN12_HIGH_L1C DATASET

Method	IoU				mIoU	Acc				mAcc	Dice				mDice	aAcc
	CRS	TNC	TKC	CDS		CRS	TNC	TKC	CDS		CRS	TNC	TKC	CDS		
SCNN [28]	56.39	00.00	34.60	00.00	22.75	89.69	00.00	45.08	00.00	33.69	72.11	00.00	51.41	00.00	30.88	60.19
KappaMask [29]	76.84	14.75	73.51	00.00	41.27	**97.54**	27.78	77.73	00.00	50.76	86.91	25.70	84.73	00.00	49.33	76.27
MCDNet [26]	68.39	13.02	65.26	32.52	44.80	77.85	18.39	84.49	59.20	59.99	81.23	23.04	78.98	49.08	58.08	72.68
CDNetv1 [12]	80.48	36.50	79.60	44.83	60.35	90.60	50.19	88.34	60.72	72.46	89.19	53.48	88.64	61.91	73.30	83.48
DBNet [27]	85.20	43.02	81.40	52.44	65.52	95.56	53.15	89.40	62.47	75.15	92.01	60.16	89.75	68.80	77.68	86.83
CDNetv2 [13]	84.98	43.37	80.67	53.40	65.60	94.80	53.13	89.32	65.34	75.65	91.88	60.50	89.30	69.62	77.83	86.68
HRCloudNet [30]	86.01	45.88	83.54	57.61	68.26	94.49	59.47	91.39	67.19	78.13	92.48	62.90	91.03	73.10	79.88	87.86
UNetMobv2 [56]	88.51	50.12	84.79	63.17	71.65	95.77	61.87	91.44	75.13	81.05	93.90	66.78	91.77	77.43	82.47	89.52
Cloud-Adapter (Ours)	**89.19**	**56.15**	**85.46**	**65.93**	**74.18**	94.08	**75.23**	**91.72**	**78.15**	**84.79**	**94.29**	**71.91**	**92.16**	**79.47**	**84.46**	**90.19**

TABLE XII

CLOUD SEGMENTATION PERFORMANCE COMPARISON OF DIFFERENT METHODS ON THE CLOUDSEN12_HIGH_L2A DATASET

Method	IoU				mIoU	Acc				mAcc	Dice				mDice	aAcc
	CRS	TNC	TKC	CDS		CRS	TNC	TKC	CDS		CRS	TNC	TKC	CDS		
SCNN [28]	62.20	00.00	52.74	00.08	28.76	85.73	00.00	76.19	00.08	40.50	76.70	00.00	69.06	00.16	36.48	67.14
KappaMask [29]	79.58	23.35	78.20	00.00	45.28	**97.24**	38.05	86.58	00.00	55.47	88.63	37.85	87.77	00.00	53.56	79.60
MCDNet [26]	72.46	14.96	67.04	31.62	46.52	85.76	21.09	85.29	42.01	58.54	84.03	26.02	80.27	48.04	59.59	75.67
CDNetv1 [12]	82.58	38.61	78.72	49.64	62.39	92.73	52.65	88.12	60.50	73.50	90.46	55.71	88.09	66.35	75.15	84.74
DBNet [27]	85.42	42.76	80.80	53.62	65.65	94.83	53.53	90.27	63.45	75.52	92.14	59.90	89.38	69.81	77.81	86.82
CDNetv2 [13]	85.35	44.01	80.85	54.00	66.05	95.65	56.37	88.06	63.56	75.91	92.09	61.12	89.41	70.13	78.19	86.88
HRCloudNet [30]	87.24	44.18	82.78	59.20	68.35	95.41	51.00	93.17	69.83	77.35	93.19	61.28	90.58	74.37	79.85	88.35
UNetMobv2 [56]	87.76	47.38	84.16	62.13	70.36	95.98	59.60	90.94	71.75	79.57	93.48	64.29	91.40	76.64	81.45	88.96
Cloud-Adapter (Ours)	**89.04**	**54.97**	**84.91**	**64.60**	**73.38**	94.65	**73.64**	**90.61**	**76.81**	**83.93**	**94.20**	**70.95**	**91.84**	**78.49**	**83.87**	**89.90**

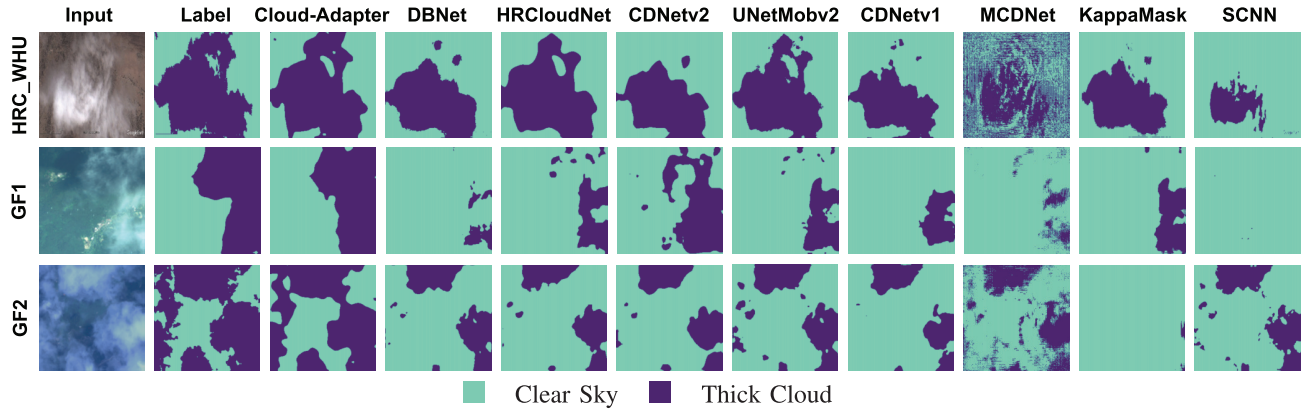


Fig. 7. Comparison of visualized segmentation results of different models on the coarse-grained cloud segmentation dataset.

TABLE XIII

CLOUD SEGMENTATION PERFORMANCE COMPARISON OF DIFFERENT METHODS ON THE L8_BIOME DATASET

Method	IoU				mIoU	Acc				mAcc	Dice				mDice	aAcc
	CRS	TNC	TKC	CDS		CRS	TNC	TKC	CDS		CRS	TNC	TKC	CDS		
MCDNet [26]	65.83	08.40	60.97	00.21	33.85	73.80	10.26	95.01	00.22	44.82	79.40	15.49	75.75	00.41	42.76	69.75
SCNN [28]	66.30	00.00	63.24	00.00	32.38	81.11	00.00	**95.12**	00.00	44.06	79.73	00.00	77.48	00.00	39.30	71.22
CDNetv1 [12]	60.69	19.63	56.79	01.21	34.58	88.19	25.51	67.43	01.22	45.59	75.54	32.81	72.44	02.40	45.80	68.16
KappaMask [29]	77.72	22.63	68.12	00.00	42.12	93.08	32.87	80.96	00.00	51.73	87.45	36.91	81.04	00.00	51.35	76.63
UNetMobv2 [56]	79.18	28.35	80.46	03.05	47.76	93.62	34.63	93.59	03.31	56.29	88.37	44.18	89.16	05.92	56.91	82.00
CDNetv2 [13]	73.81	24.85	75.85	00.00	43.63	90.18	31.95	89.81	00.00	52.98	84.93	39.81	86.27	00.00	52.75	78.56
HRCloudNet [30]	72.41	30.89	70.76	00.00	43.51	85.50	43.70	85.88	00.00	53.77	83.99	47.20	82.88	00.00	53.52	77.04
DBNet [27]	80.19	38.18	**82.37**	04.90	51.41	**96.51**	49.34	87.89	05.03	59.70	89.00	55.25	**90.34**	09.34	60.99	**83.62**
Cloud-Adapter (Ours)	**85.81**	**44.41**	70.75	**29.13**	**57.53**	90.63	**78.76**	75.35	**32.65**	**69.34**	**92.36**	**61.51**	82.87	**45.12**	**70.46**	81.79

b) Visualization comparison: Fig. 7 compares the visualized segmentation results for three binary-class datasets. Cloud-Adapter stands out for its ability to accurately capture cloud boundaries and reduce oversegmentation issues, which are common in other methods, such as DBNet and HRCloudNet. The results illustrate its superiority in segmenting clear sky and cloud regions, even under complex conditions.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:19:28 UTC from IEEE Xplore. Restrictions apply.

TABLE XIV
COMPARISON OF CLOUD SEGMENTATION PERFORMANCE ACROSS DIFFERENT LAND COVER TYPES IN THE L8_BIOME DATASET

Scene	Method	mIoU	mAcc	mDice	aAcc
Grass/Crops	KappaMask [29]	36.29	62.25	42.17	76.56
	CDNetv1 [12]	39.99	59.01	44.22	73.04
	CDNetv2 [13]	55.17	71.84	59.23	86.78
	HRCloudNet [30]	50.47	66.21	56.19	81.99
	MCDNet [26]	28.92	69.85	32.92	81.61
	SCNN [28]	43.81	70.48	46.86	83.13
	DBNet [27]	**59.08**	73.85	**63.31**	**89.85**
	UNetMobv2 [56]	49.46	**74.78**	54.39	88.21
	Cloud-Adapter (Ours)	41.60	63.10	48.12	72.71
Urban	KappaMask [29]	48.83	73.39	55.55	86.78
	CDNetv1 [12]	46.30	64.64	51.03	81.44
	CDNetv2 [13]	56.27	74.55	61.74	89.13
	HRCloudNet [30]	56.47	71.57	62.85	87.99
	MCDNet [26]	31.90	70.68	36.48	83.87
	SCNN [28]	48.63	71.55	52.05	85.66
	DBNet [27]	**59.68**	76.73	**65.25**	**92.14**
	UNetMobv2 [56]	50.24	**79.90**	55.39	91.59
	Cloud-Adapter (Ours)	51.76	76.84	58.36	86.27
Wetlands	KappaMask [29]	40.38	61.72	47.59	75.12
	CDNetv1 [12]	34.50	56.45	39.69	66.57
	CDNetv2 [13]	40.54	59.62	46.71	74.78
	HRCloudNet [30]	41.48	57.89	47.93	76.38
	MCDNet [26]	26.23	55.86	30.75	68.51
	SCNN [28]	34.38	55.90	37.76	67.42
	DBNet [27]	49.79	67.54	56.52	83.53
	UNetMobv2 [56]	41.54	64.58	47.82	77.59
	Cloud-Adapter (Ours)	**50.16**	**74.26**	**57.09**	**86.08**
Snow/Ice	KappaMask [29]	24.41	43.24	29.37	56.74
	CDNetv1 [12]	20.93	38.00	23.82	44.90
	CDNetv2 [13]	24.99	41.18	28.51	55.37
	HRCloudNet [30]	22.20	36.91	25.41	47.58
	MCDNet [26]	10.43	25.64	13.30	29.59
	SCNN [28]	12.43	23.33	14.37	24.70
	DBNet [27]	29.72	47.25	34.04	67.69
	UNetMobv2 [56]	23.50	41.99	27.96	56.48
	Cloud-Adapter (Ours)	**32.01**	**56.99**	**36.25**	**73.15**
Barren	KappaMask [29]	38.36	60.64	45.31	76.96
	CDNetv1 [12]	31.50	48.05	36.15	59.96
	CDNetv2 [13]	41.24	59.24	46.03	76.33
	HRCloudNet [30]	44.33	60.53	50.19	77.51
	MCDNet [26]	26.56	60.52	31.34	73.00
	SCNN [28]	36.93	60.42	40.55	73.35
	DBNet [27]	47.41	65.13	53.05	80.38
	UNetMobv2 [56]	43.66	67.80	49.67	**84.01**
	Cloud-Adapter (Ours)	**48.95**	**72.35**	**56.45**	83.49
Forest	KappaMask [29]	38.32	57.00	44.46	80.07
	CDNetv1 [12]	33.86	48.50	38.67	71.81
	CDNetv2 [13]	41.94	58.62	47.87	81.34
	HRCloudNet [30]	42.93	58.30	49.64	81.76
	MCDNet [26]	30.05	57.39	35.23	77.71
	SCNN [28]	37.62	55.44	41.39	77.92
	DBNet [27]	44.93	62.40	50.81	85.22
	UNetMobv2 [56]	45.23	64.89	51.17	**86.89**
	Cloud-Adapter (Ours)	**48.09**	**68.76**	**56.25**	82.30
Shrubland	KappaMask [29]	40.87	59.71	48.28	77.26
	CDNetv1 [12]	32.70	49.79	38.89	61.35
	CDNetv2 [13]	42.04	59.48	48.51	76.94
	HRCloudNet [30]	41.68	57.22	48.98	76.79
	MCDNet [26]	28.66	57.34	34.20	71.57
	SCNN [28]	37.11	56.93	41.61	71.25
	DBNet [27]	46.49	63.41	53.76	80.88
	UNetMobv2 [56]	41.58	63.24	48.19	79.79
	Cloud-Adapter (Ours)	**51.18**	**72.01**	**59.19**	**87.42**
Water	KappaMask [29]	42.42	64.40	49.36	81.19
	CDNetv1 [12]	44.43	61.84	49.20	82.83
	CDNetv2 [13]	48.72	64.17	54.10	85.07
	HRCloudNet [30]	46.74	60.46	52.96	82.62
	MCDNet [26]	29.43	60.95	33.81	79.71
	SCNN [28]	41.82	61.56	45.51	80.86
	DBNet [27]	**52.42**	67.85	**58.29**	87.15
	UNetMobv2 [56]	49.58	**70.25**	55.37	**88.24**
	Cloud-Adapter (Ours)	46.43	67.75	53.26	82.05

2) Multiclass Cloud Segmentation:

a) Quantitative comparison: Cloud-Adapter achieves state-of-the-art performance on multiclass cloud segmentation datasets, which include four categories: clear sky (CRS), thin clouds (TNC), thick clouds (TKC), and cloud shadows (CDS). On the CloudSEN12_High_L1C dataset (Table XI), Cloud-Adapter achieves an mIoU of 74.18%, with IoUs of 89.19% (CRS), 85.46% (TKC), 56.15% (TNC), and 65.93% (CDS). It outperforms methods like UNetMobv2 and HRCloudNet, which show lower scores, particularly in challenging classes such as TNC and CDS. On the CloudSEN12_High_L2A dataset (Table XII), Cloud-Adapter achieves an mIoU of 73.38%, with IoUs of 89.04% (CRS), 84.91% (TKC), 54.97% (TNC), and 64.60% (CDS). Its mDice score of 83.87% further highlights its balanced performance across all categories. On the L8_Biome dataset (Tables XIII and XIV), Cloud-Adapter attains the highest mIoU (57.53%), mAcc (69.34%), mDice (70.46%), and aAcc (81.79%). It excels in challenging categories such as snow/ice (32.01%) and shrubland (51.18%) and demonstrates strong overall performance across diverse land scenes, often outperforming competitors like DBNet and UNetMobv2.

b) Visualization comparison: Figs. 8 and 9 present Cloud-Adapter's segmentation results on multiclass datasets (CloudSEN12_High_L1C and CloudSEN12_High_L2A), accurately distinguishing thick clouds, thin clouds, and cloud shadows in complex conditions. The figures also demonstrate its performance across varying cloud cover levels: "Low" (less than 35%), "Medium" (35%–65%), and "High" (over 65%). These results highlight Cloud-Adapter's robustness and adaptability to diverse atmospheric scenarios. Fig. 10 highlights Cloud-Adapter's segmentation performance on the L8_Biome dataset, demonstrating superior generalization across diverse biomes, such as barren, forest, and snow/ice regions. These results further emphasize the model's capability to adapt to varying environmental conditions while maintaining high accuracy and precision, solidifying its effectiveness for complex cloud segmentation tasks in remote sensing applications.

3) Domain Generalized Cloud Segmentation: To further evaluate the robustness of Cloud-Adapter, we conducted supplementary experiments under two representative domain shift scenarios: 1) training on GF12MS_WHU_GF2 and testing on HRC_WHU and 2) training on CloudSEN12_High_L1C

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:19:28 UTC from IEEE Xplore. Restrictions apply.

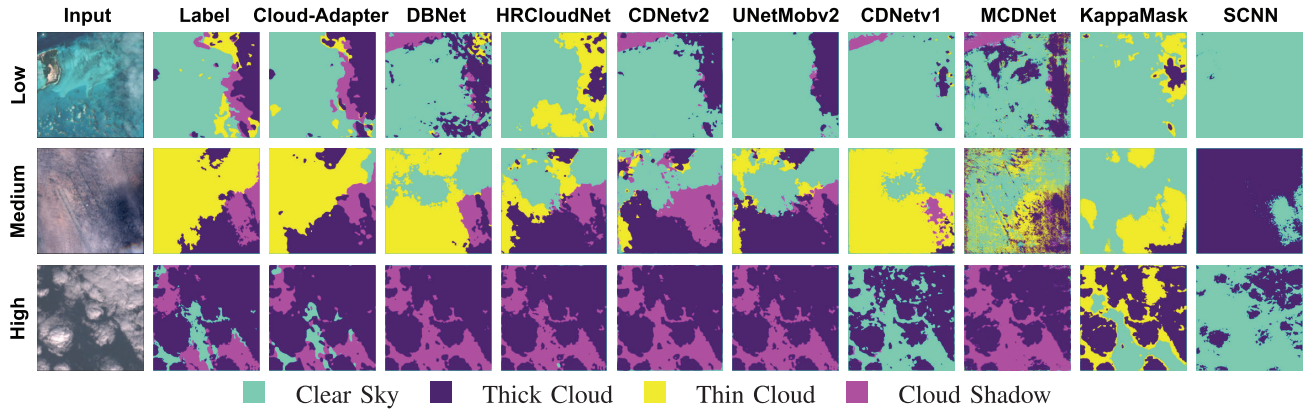


Fig. 8. Comparison of visualized segmentation results of different models on the CloudSEN12_High_L1C dataset.

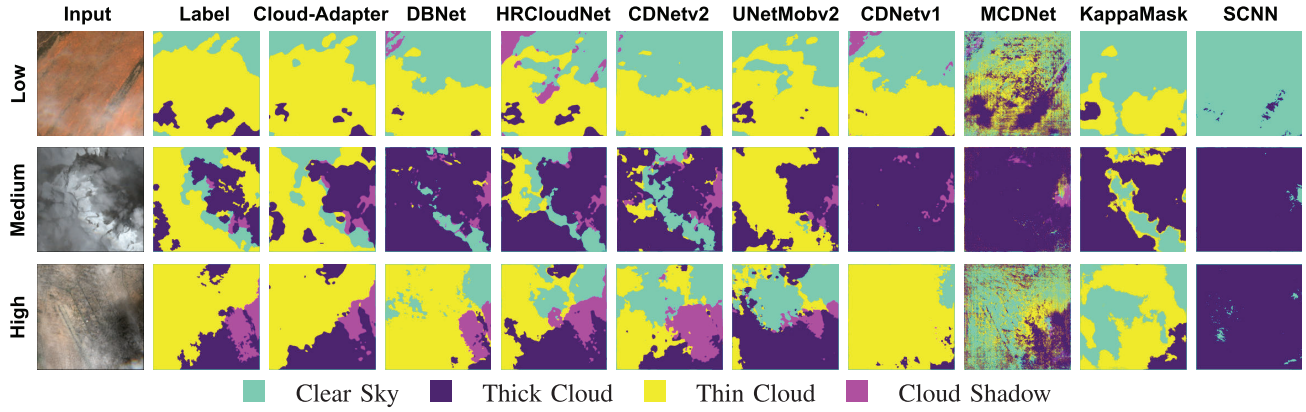


Fig. 9. Comparison of visualized segmentation results of different models on the CloudSEN12_High_L2A dataset.

TABLE XV
COMPARISON OF DIFFERENT MODELS IN TERMS OF GENERALIZATION ABILITY UNDER CROSS-DATASET SETTINGS

GF12MS_WHU_GF2→HRC_WHU					CloudSEN12_High_L1C→CloudSEN12_High_L2A				
Methods	**aAcc**	**mIoU**	**mAcc**	**mDice**	**Methods**	**aAcc**	**mIoU**	**mAcc**	**mDice**
KappaMask [29]	53.53	36.37	67.05	53.25	SCNN [28]	60.49	22.44	33.16	30.52
CDNetv1 [12]	57.77	39.53	64.95	56.18	KappaMask [29]	75.27	38.84	47.82	46.83
HRCloudNet [30]	78.23	52.54	64.05	65.68	MCDNet [26]	74.95	42.02	52.70	52.92
MCDNet [26]	69.95	52.02	77.31	67.95	CDNetv1 [12]	82.15	54.01	63.14	66.33
UNetMobv2 [56]	77.98	58.43	74.86	72.61	DBNet [27]	84.31	58.03	66.66	70.35
DBNet [27]	81.19	65.04	84.58	78.30	CDNetv2 [13]	85.01	61.23	71.03	73.86
SCNN [28]	83.06	67.54	85.82	80.14	HRCloudNet [30]	85.78	63.06	72.63	75.53
CDNetv2 [13]	85.37	70.95	87.85	82.60	UNetMobv2 [56]	87.26	65.16	73.54	76.89
Cloud-Adapter (Ours)	**89.73**	**80.18**	**87.89**	**88.92**	Cloud-Adapter (Ours)	**89.40**	**72.52**	**83.43**	**83.22**

and testing on CloudSEN12_High_L2A. As shown in Table XV, Cloud-Adapter consistently achieves superior performance across both settings, attaining 80.18% mIoU and 88.92% mDice in the GF12MS_WHU_GF2 →HRC_WHU transfer, and 72.52% mIoU and 83.22% mDice in the CloudSEN12_High_L1C→L2A case. Compared with all baseline methods, Cloud-Adapter exhibits clear advantages in segmentation performance, providing additional and compelling evidence for its strong cross-domain generalization capability.

TABLE XVI
PERFORMANCE AND PARAMETER EFFICIENCY COMPARISON BETWEEN CLOUD-ADAPTER AND RECENT PARAMETER-EFFICIENT ADAPTATION METHODS ON THE CLOUDSEN12_HIGH_L1C DATASET

Method	**mIoU**	**mAcc**	**mDice**	**aAcc**	***Params (M)**
Rein [62]	73.56	**84.87**	84.03	89.77	2.99
ViT-Adapter [49]	72.56	83.28	83.71	88.74	47.80
Cloud-Adapter (Ours)	**74.18**	84.79	**84.46**	**90.19**	**1.82**

4) Parameter-Efficient VFM Adaptation for Cloud Segmentation: To the best of our knowledge, Cloud-Adapter is the first approach to introduce parameter-efficient fine-tuning of VFMs for remote sensing, with a particular focus on cloud segmentation tasks. Compared to existing parameter-efficient adaptation methods, Cloud-Adapter provides notable innovations and practical advantages. For example, ViT-Adapter [49] employs generic adapter modules for dense prediction tasks, but at the cost of a large number of additional parameters (47.80 M), making it less suitable for resource-

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:19:28 UTC from IEEE Xplore. Restrictions apply.

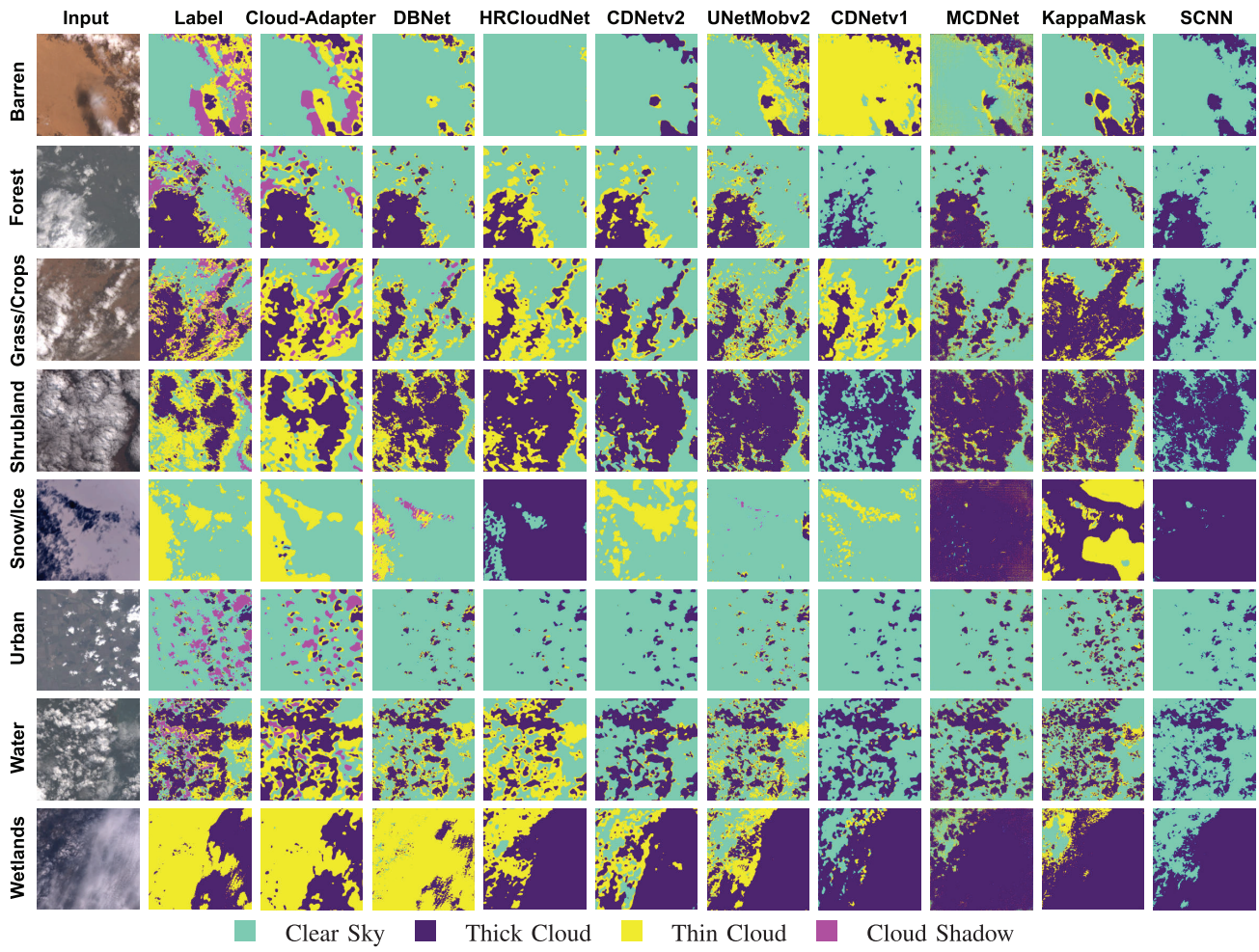


Fig. 10. Comparison of visualized segmentation results of different models on the L8_Biome dataset.

constrained or large-scale remote sensing applications. Rein [62], on the other hand, adopts a more complex architecture by introducing learnable instance tokens to enhance domain generalization, which increases model complexity and requires careful token design. In contrast, Cloud-Adapter is specifically designed to address the highly variable spatial distribution, diverse scales, and inconsistent densities of clouds in remote sensing imagery. It incorporates an SPM to extract dense, multiscale contextual cues, enabling the frozen VFM backbone to adapt more effectively to these unique characteristics. As shown in Table XVI, Cloud-Adapter achieves an mIoU of 74.18, which is +0.62 and +1.62 higher than Rein and ViT-Adapter, respectively, while requiring only 1.82 M additional trainable parameters. These results demonstrate that Cloud-Adapter not only advances parameter-efficient adaptation in remote sensing but also brings new perspectives and significant practical benefits for challenging cloud segmentation scenarios.

V. Conclusion

In this article, we introduced Cloud-Adapter, a parameter-efficient fine-tuning method for cloud segmentation in remote sensing images. By leveraging a frozen VFM and adding a lightweight adapter, Cloud-Adapter reduces the number of trainable parameters while achieving SOTA performance across various datasets. Our approach provides an effective solution for robust cloud segmentation, eliminating the need for extensive retraining and ensuring both efficiency and adaptability.

A. Limitations

The performance of Cloud-Adapter depends heavily on the VFM, and the large parameters of VFMs make it difficult to deploy on edge devices with limited resources. Additionally, in the complex dataset (L8_Biome), which encompasses diverse land cover scenarios, our method achieved a leading mIoU score of 57.53%. However, a substantial gap remains when compared to manually annotated masks. This highlights the potential for improving algorithm efficiency and developing new techniques to enhance segmentation accuracy.

References

[1] C. E. Bulgin, R. I. Maidment, D. Ghent, and C. J. Merchant, "Stability of cloud detection methods for Land Surface Temperature (LST) climate data records (CDRs)," *Remote Sens. Environ.*, vol. 315, Dec. 2024, Art. no. 114440.

[2] C.-C. Young, Y.-C. Cheng, M.-A. Lee, and J.-H. Wu, "Accurate reconstruction of satellite-derived SST under cloud and cloud-free areas using a physically-informed machine learning approach," *Remote Sens. Environ.*, vol. 313, Nov. 2024, Art. no. 114339.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:19:28 UTC from IEEE Xplore. Restrictions apply.

[3] Z. Zhu and C. E. Woodcock, "Automated cloud, cloud shadow, and snow detection in multitemporal Landsat data: An algorithm designed specifically for monitoring land cover change," *Remote Sens. Environ.*, vol. 152, pp. 217–234, Sep. 2014.

[4] Y. Zhang, B. Guindon, and J. Cihlar, "An image transform to characterize and compensate for spatial variations in thin cloud contamination of Landsat images," *Remote Sens. Environ.*, vol. 82, nos. 2–3, pp. 173–187, Oct. 2002.

[5] S. Chen, X. Chen, J. Chen, P. Jia, X. Cao, and C. Liu, "An iterative haze optimized transformation for automatic cloud/haze detection of Landsat imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 5, pp. 2682–2694, May 2016.

[6] Z. Zhu, S. Wang, and C. E. Woodcock, "Improvement and expansion of the fmask algorithm: Cloud, cloud shadow, and snow detection for Landsats 4–7, 8, and sentinel 2 images," *Remote Sens. Environ.*, vol. 159, pp. 269–277, Mar. 2015.

[7] S. Foga et al., "Cloud detection algorithm comparison and validation for operational Landsat data products," *Remote Sens. Environ.*, vol. 194, pp. 379–390, Jun. 2017.

[8] Z. Li, H. Shen, Q. Cheng, Y. Liu, S. You, and Z. He, "Deep learning based cloud detection for medium and high resolution remote sensing images of different sensors," *ISPRS J. Photogramm. Remote Sens.*, vol. 150, pp. 197–212, Apr. 2019.

[9] J. H. Jeppesen, R. H. Jacobsen, F. Inceoglu, and T. S. Toftegaard, "A cloud detection algorithm for satellite imagery based on deep learning," *Remote Sens. Environ.*, vol. 229, pp. 247–259, Aug. 2019.

[10] Z. Shao, Y. Pan, C. Diao, and J. Cai, "Cloud detection in remote sensing images based on multiscale features-convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 6, pp. 4062–4076, Jun. 2019.

[11] M. Shi, F. Xie, Y. Zi, and J. Yin, "Cloud detection of remote sensing images by deep learning," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2016, pp. 701–704.

[12] J. Yang, J. Guo, H. Yue, Z. Liu, H. Hu, and K. Li, "CDnet: CNN-based cloud detection for remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 6195–6211, Aug. 2019.

[13] J. Guo, J. Yang, H. Yue, H. Tan, C. Hou, and K. Li, "CDnetV2: CNN-based cloud detection for remote sensing imagery with cloud-snow coexistence," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 700–713, Jan. 2021.

[14] Q. He, X. Sun, Z. Yan, and K. Fu, "DABNet: Deformable contextual and boundary-weighted network for cloud detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5601216.

[15] A. Dosovitskiy et al., "An image is worth 16*x*16 words: Transformers for image recognition at scale," in *Proc. ICLR*, 2020, pp. 1–21.

[16] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 10012–10022.

[17] K. He, X. Chen, S. Xie, Y. Li, P. Dollár, and R. Girshick, "Masked autoencoders are scalable vision learners," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 16000–16009.

[18] A. Kirillov et al., "Segment anything," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2023, pp. 4015–4026.

[19] M. Oquab et al., "DINOv2: Learning robust visual features without supervision," *Trans. Mach. Learn. Res. J.*, vol. 2023, pp. 1–31, Apr. 2023.

[20] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "ImageNet: A large-scale hierarchical image database," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Miami, FL, USA, 2009, pp. 248–255.

[21] C.-F. R. Chen, Q. Fan, and R. Panda, "CrossViT: Cross-attention multi-scale vision transformer for image classification," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, 2021, pp. 357–366.

[22] H. Chen, P. Wei, G. Guo, and S. Gao, "SAM-COD: SAM-guided unified framework for weakly-supervised camouflaged object detection," in *Proc. ECCV*, Mar. 2025, pp. 315–331.

[23] X. He, Y. Zhou, J. Zhao, D. Zhang, R. Yao, and Y. Xue, "Swin transformer embedding UNet for remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4408715.

[24] F. Li et al., "Mask DINO: Towards a unified transformer-based framework for object detection and segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 3041–3050.

[25] G. Li, H. Zheng, D. Liu, C. Wang, B. Su, and C. Zheng, "SemMAE: Semantic-guided masking for learning masked autoencoders," in *Proc. NeurIPS*, 2022, pp. 14290–14302.

[26] J. Dong, Y. Wang, Y. Yang, M. Yang, and J. Chen, "MCDNet: Multilevel cloud detection network for remote sensing images based on dual-perspective change-guided and multi-scale feature fusion," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 129, May 2024, Art. no. 103820.

[27] C. Lu, M. Xia, M. Qian, and B. Chen, "Dual-branch network for cloud and cloud shadow segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5410012.

[28] D. Chai, J. Huang, M. Wu, X. Yang, and R. Wang, "Remote sensing image cloud detection using a shallow convolutional neural network," *ISPRS J. Photogramm. Remote Sens.*, vol. 209, pp. 66–84, Mar. 2024.

[29] M. Domnich et al., "KappaMask: AI-based cloudmask processor for Sentinel-2," *Remote Sens.*, vol. 13, no. 20, p. 4100, Oct. 2021.

[30] J. Li et al., "High-resolution cloud detection network," *J. Electron. Imag.*, vol. 33, no. 4, Jul. 2024, Art. no. 043027.

[31] S. Zhu, Z. Li, and H. Shen, "Transferring deep models for cloud detection in multisensor images via weakly supervised learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5609518.

[32] Q. Yan et al., "Toward high-quality HDR deghosting with conditional diffusion models," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 5, pp. 4011–4026, May 2024.

[33] Q. Yan et al., "From dynamic to static: Stepwisely generate HDR image for ghost removal," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 35, no. 2, pp. 1409–1421, Feb. 2025.

[34] Y. Zhang, H. Zhang, N. M. Nasrabadi, and T. S. Huang, "Multi-metric learning for multi-sensor fusion based classification," *Inf. Fusion*, vol. 14, no. 4, pp. 431–440, Oct. 2013.

[35] Q. Yan et al., "Uncertainty estimation in HDR imaging with Bayesian neural networks," *Pattern Recognit.*, vol. 156, Dec. 2024, Art. no. 110802.

[36] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. 18th Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, vol. 9351. Cham, Switzerland: Springer, 2015, pp. 234–241.

[37] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[38] T.-Y. Lin, P. Dollár, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 936–944.

[39] L.-C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, and A. L. Yuille, "DeepLab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected CRFs," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 4, pp. 834–848, Apr. 2018.

[40] J. Devlin, M. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proc. NAACL*, 2018, p. 2.

[41] T. B. Brown et al., "Language models are few-shot learners," in *Proc. NeurIPS*, 2020, pp. 1877–1901.

[42] A. Radford et al., "Learning transferable visual models from natural language supervision," in *Proc. ICML*, 2021, pp. 8748–8763.

[43] S. J. Pan and Q. Yang, "A survey on transfer learning," *IEEE Trans. Knowl. data Eng.*, vol. 22, no. 10, pp. 1345–1359, Oct. 2009.

[44] Y. Ganin and V. Lempitsky, "Unsupervised domain adaptation by backpropagation," in *Proc. ICML*, 2014, pp. 1180–1189.

[45] N. Houlsby et al., "Parameter-efficient transfer learning for NLP," in *Proc. ICML*, 2019, pp. 2790–2799.

[46] J. E. Hu et al., "LoRA: Low-rank adaptation of large language models," in *Proc. ICLR*, 2021, pp. 1–13.

[47] H. Ye, J. Zhang, S. Liu, X. Han, and W. Yang, "IP-adapter: Text compatible image prompt adapter for text-to-image diffusion models," 2023, *arXiv:2308.06721*.

[48] C. Mou et al., "T2I-adapter: Learning adapters to dig out more controllable ability for text-to-image diffusion models," in *Proc. AAAI Conf. Artif. Intell.*, Mar. 2024, vol. 38, no. 5, pp. 4296–4304.

[49] Z. Chen et al., "Vision transformer adapter for dense predictions," in *Proc. ICLR*, 2022, pp. 1–20.

[50] T. Chen et al., "SAM-adapter: Adapting segment anything in underperformed scenes," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshops (ICCVW)*, Oct. 2023, pp. 3367–3375.

[51] F. Chollet, "Xception: Deep learning with depthwise separable convolutions," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 1800–1807.

[52] H. Lin, X. Cheng, X. Wu, and D. Shen, "CAT: Cross attention in vision transformer," in *Proc. IEEE Int. Conf. Multimedia Expo. (ICME)*, Jul. 2022, pp. 1–6.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:19:28 UTC from IEEE Xplore. Restrictions apply.

[53] Z. Huang, X. Wang, L. Huang, C. Huang, Y. Wei, and W. Liu, "CCNet: Criss-cross attention for semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 603–612.
[54] O. Petit, N. Thome, C. Rambour, L. Themyr, T. Collins, and L. Soler, "U-Net transformer: Self and cross attention for medical image segmentation," in *Proc. Int. Workshop Mach. Learn. Med. Imag.* Cham, Switzerland: Springer, 2021, pp. 267–276.
[55] X. Zou, K. Li, J. Xing, P. Tao, and Y. Cui, "PMAA: A progressive multi-scale attention autoencoder model for high-performance cloud removal from multi-temporal satellite imagery," in *Proc. ECAI*, 2023, pp. 3165–3172.
[56] C. Aybar et al., "CloudSEN12, a global dataset for semantic understanding of cloud and cloud shadow in Sentinel-2," *Sci. Data*, vol. 9, no. 1, p. 782, Dec. 2022.
[57] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *Proc. ICLR*, 2018, pp. 1–18.
[58] K. He, X. Zhang, S. Ren, and J. Sun, "Delving deep into rectifiers: Surpassing human-level performance on ImageNet classification," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1026–1034.
[59] B. Cheng, I. Misra, A. G. Schwing, A. Kirillov, and R. Girdhar, "Masked-attention mask transformer for universal image segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 1290–1299.
[60] Z. Liu, H. Mao, C.-Y. Wu, C. Feichtenhofer, T. Darrell, and S. Xie, "A ConvNet for the 2020s," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 11976–11986.
[61] C. Zhou, C. C. Loy, and B. Dai, "Extract free dense labels from CLIP," in *Proc. ECCV*, 2022, pp. 696–712.
[62] Z. Wei et al., "Stronger, fewer & superior: Harnessing vision foundation models for domain generalized semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 28619–28630.



Xuechao Zou received the B.E. degree and the M.S. degree from the School of Computer Technology and Application, Qinghai University, Xining, China, in 2021 and 2024, respectively. He is currently pursuing the Ph.D. degree with the School of Computer Science and Technology, Beijing Jiaotong University, Beijing, China.

His research interests include computer vision, particularly remote sensing image processing.



Shun Zhang is currently pursuing the B.E. degree with the School of Computer Technology and Application, Qinghai University, Xining, China.

His research interests include computer vision, particularly in the field of remote sensing image processing.



Kai Li (Student Member, IEEE) received the B.E. degree from the School of Computer Technology and Application, Qinghai University, Xining, China, in 2020, and the M.S. degree from the Department of Computer Science and Technology, Tsinghua University, Beijing, China, in 2024, where he is currently pursuing the Ph.D. degree with the Department of Computer Science and Technology.

His research interests include speech separation and audio–visual learning.



Shiying Wang received the B.E. degree from Jilin Agricultural University, Changchun, China, in 2015, and the M.S. degree from the School of Computer Technology and Application, Qinghai University, Xining, China, in 2018, where she is currently pursuing the Ph.D. degree.

Her research interests include remote sensing image processing, particularly in grassland informatics.



Junliang Xing (Senior Member, IEEE) received the dual B.S. degree in computer science and mathematics from Xi'an Jiaotong University, Shaanxi, China, in 2007, and the Ph.D. degree in computer science and technology from Tsinghua University, Beijing, China, in 2012.

He is currently a Professor with the Department of Computer Science and Technology, Tsinghua University. He has published over 150 peer-reviewed papers with more than 22000 citations from Google Scholar. His current research interests include computer vision and gaming problems related to single-agent and multiagent learning, as well as human–computer interactive learning.



Lei Jin received the bachelor's degree from Beijing University of Posts and Telecommunications (BUPT), Beijing, China, in 2015.

He is currently an Associate Research Fellow with BUPT. His research interests include machine learning and pattern recognition, with a focus on 6-D and human pose estimation. Moreover, he concentrated on network security and analyzed traffic security while pursuing his Ph.D. degree.



Congyan Lang (Member, IEEE) received the Ph.D. degree from Beijing Key Laboratory of Traffic Data Analysis and Mining, School of Computer and Information Technology, Beijing Jiaotong University, Beijing, China, in 2006.

She was a Visiting Professor with the Department of Electrical and Computer Engineering, National University of Singapore, Singapore, from 2010 to 2011. From 2014 to 2015, she was a Visiting Professor with the Department of Computer Science, University of Rochester, Rochester, NY, USA. She is a Professor with the School of Computer and Information Technology, Beijing Jiaotong University. She has published over 80 research articles in various journals and refereed conferences. Her research areas include computer vision and machine learning.



Pin Tao (Member, IEEE) received the B.S. and Ph.D. degrees in computer science and technology from Tsinghua University, Beijing, China, in 1997 and 2002, respectively.

He is an Associate Professor with the Computer Science and Technology Department, Tsinghua University. He is also the Vice Director of the Key Laboratory of Pervasive Computing, Ministry of Education. He has published over 80 papers and holds more than ten patents. His current research interests mainly focus on human–AI hybrid intelligence and multimedia-embedded processing.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:19:28 UTC from IEEE Xplore. Restrictions apply.